Rosenberg's conjecture in arXiv:1302.2286 was answered by arXiv:1509.08318

Literature status packet

2026-06-21

Status

literature_already_answered. The source paper records Rosenberg's conjecture as an open problem. The conjecture was later explicitly confirmed by Tikuisis–White–Winter.

Source passage

The source/open-problem paper is Ben Hayes, *An l^p-Version of von Neumann Dimension for Banach Space Representations of Sofic Groups II*, arXiv:1302.2286 [1].

On source PDF page 32, after Theorem 3.4, Hayes writes that by Brown–Ozawa and Choi–Effros, if every amenable group were in the class $\mathcal{C}$, then $C^*_\lambda(\Gamma)$ would be quasidiagonal for every amenable group. The paper then says that this is Rosenberg's conjecture and that it is an open problem of major current interest.

Supporting answer

The supporting paper is Aaron Tikuisis, Stuart White, and Wilhelm Winter, *Quasidiagonality of nuclear C^*-algebras*, arXiv:1509.08318 [2]. Its abstract states as a consequence of the main theorem that it confirms the Rosenberg conjecture: discrete amenable groups have quasidiagonal C^*-algebras.

This directly answers the source status remark: for every discrete amenable group Γ, the reduced group algebra $C^*_\lambda(\Gamma)$ is quasidiagonal.

Scope limitations

This is a literature-status packet only. It does not answer Hayes's closing Conjectures 4.1–4.3 about l^p-dimension. It records the later resolution of the Rosenberg-conjecture open problem explicitly cited in the source paper.

References

[1] B. Hayes, *An l^p-Version of von Neumann Dimension for Banach Space Representations of Sofic Groups II*, arXiv:1302.2286, 2013.

[2] A. Tikuisis, S. White, and W. Winter, *Quasidiagonality of nuclear C*-algebras*, arXiv:1509.08318, 2015.